DETAILED DESCRIPTION OF THE INVENTION — SOFTWARE ARCHITECTURES, AI INTERPRETATION, AND ADAPTIVE MODELING SYSTEMS

The intraluminal physiologic monitoring device may operate in conjunction with software systems configured to interpret collected physiologic signals, generate derived insights, and adapt system behavior over time. Software architectures may be implemented locally on-device, within companion computing platforms, or across distributed cloud-based computational environments.

Signal processing pipelines may include preprocessing, feature extraction, normalization, temporal alignment, pattern recognition, anomaly detection, and predictive modeling. Algorithms may transform raw physiologic data into abstracted indices representing trends, states, variability patterns, or recovery characteristics without exposing explicit anatomical information.

Artificial intelligence systems may include machine learning models, neural networks, probabilistic models, reinforcement learning systems, or hybrid analytical frameworks trained using individual user data, aggregated population datasets, or simulated physiologic models. Adaptive learning processes may personalize interpretation based on longitudinal behavioral and physiologic patterns.

Software-defined sensing architectures may dynamically adjust device operation by modifying sampling frequency, sensor activation, calibration parameters, or data transmission behavior in response to detected physiologic conditions. Such adaptive control may improve measurement reliability while reducing power consumption.

Predictive modeling systems may estimate future physiologic states, detect transitions between physiologic conditions, or identify deviations from individualized baselines. AI models may continuously refine predictions through feedback loops incorporating newly acquired data.

Data visualization environments may present results using dashboards, symbolic indicators, trend curves, scores, or other non-explicit graphical representations designed to communicate insight while preserving user privacy. Visualization systems may be customized according to user preference, clinician workflows, or research protocols.

Software architectures may support interoperability with external platforms including health record systems, research databases, telehealth environments, or partner applications through standardized application programming interfaces (APIs). Secure permission frameworks may regulate data access and sharing.

Certain embodiments may include digital twin modeling in which computational representations of physiologic behavior evolve alongside measured data to simulate future responses or optimize device interaction strategies.

The software, AI interpretation, and adaptive modeling systems described herein are illustrative rather than limiting. Any computational architecture capable of deriving physiologic insight from intraluminal sensing data is considered within the scope of the present disclosure.